Learning History-Dependent Constitutive Laws From Data

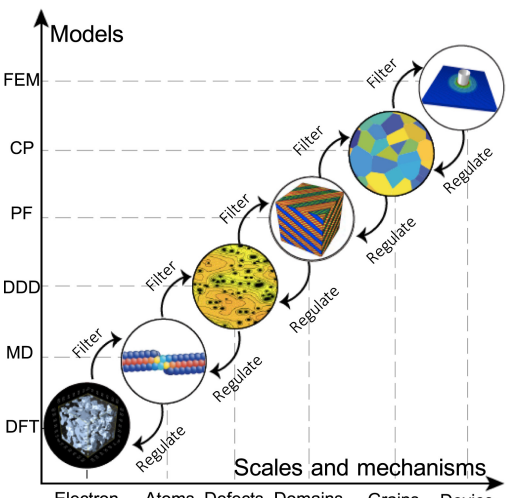
Mayank Raj

Candidacy Talk
ACM, Caltech

October 30, 2026

Introduction

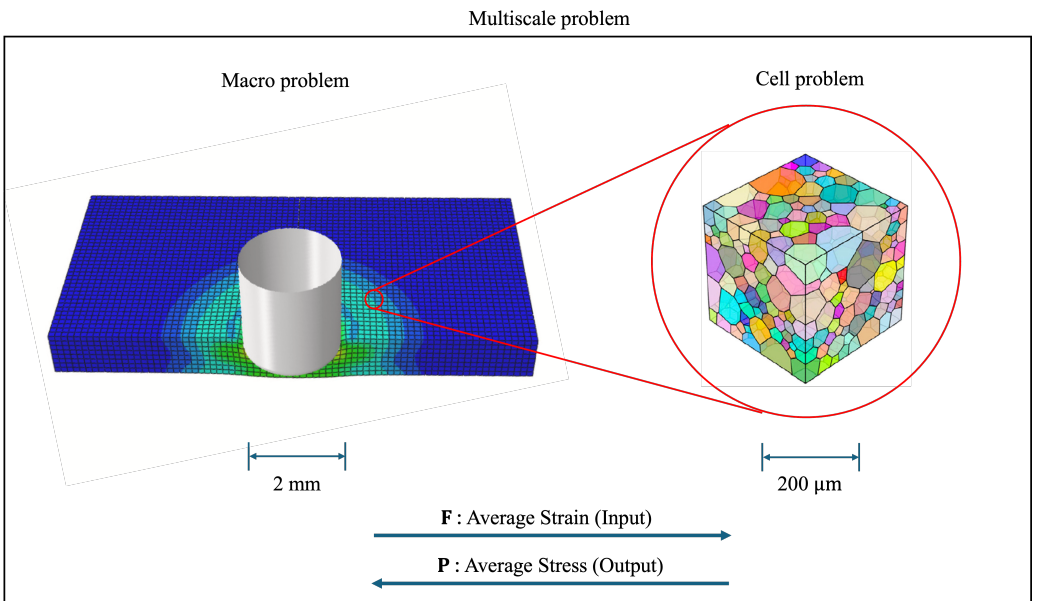
Multiscale Modeling



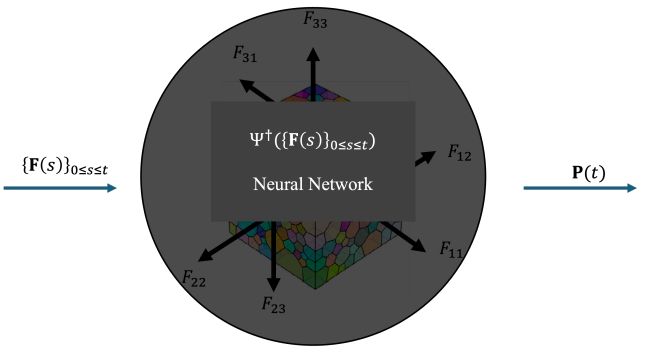
Kovachki et al., 2022

Introduction

Two-scale Homogenization



Surrogate Model of The Cell Problem



Magnesium Polycrystal, Viscoplastic
Viscoelastic

Goal

Given the dataset

$$\{(F^{(n)}(t), P^{(n)}(t))\}_{n=1,0\le t\le T}^{N},$$

recover or learn the map $\Psi^\dagger$.

Overview

- Learn a surrogate model for the cell problem
  - 3D magnesium polycrystal with fixed microstructure
  - 1D and 2D viscoelastic material with varying microstructure
  - 1D viscoplastic material with varying microstructure
- Impose desired properties on the surrogate model
  - Desired mathematical properties for the well-posedness of the macro-problem.
  - Physical properties: Dissipativity (Second Law of Thermodynamics) and limiting behavior away from the data support.
- Performance of the trained model
  - Number of internal variables
  - Predictive accuracy of the stress
- Understand the uniqueness of the surrogate model up to linear transform.
  - Internal variables are identifiable up to a linear transformation.
  - Empirical results and supporting theory.

Part I

Learning for fixed Microstructure

HYPOTHESIS MODEL CLASS FOR $\Psi^\dagger$

- Energy density function

$$\mathcal{W} : \mathbb{R}^6 \times \mathbb{R}^n \to \mathbb{R}^+ \quad \text{such that} \quad (F, \xi) \mapsto \mathcal{W}(F, \xi).$$

- Dissipation function

$$\mathcal{D} : \mathbb{R}^n \to \mathbb{R}^+ \quad \text{such that} \quad \dot{\xi} \mapsto \mathcal{D}(\dot{\xi}).$$

- Material model as ODE

$$P(t) = \partial_F \mathcal{W}(F(t), \xi(t))$$
$$\partial_{\dot{\xi}} \mathcal{D}(\dot{\xi}) + \partial_\xi \mathcal{W}(F(t), \xi(t)) = 0, \quad \xi(0) = 0.$$

Key points

- Learning $\Psi^\dagger$ is equivalent to learning $\mathcal{W}$ and $\mathcal{D}$ in the hypothesis model class.
- $\mathcal{W}$ and $\mathcal{D}$ are to be learned using samples of (F, P).
- Samples of internal variables ξ are not available for learning.

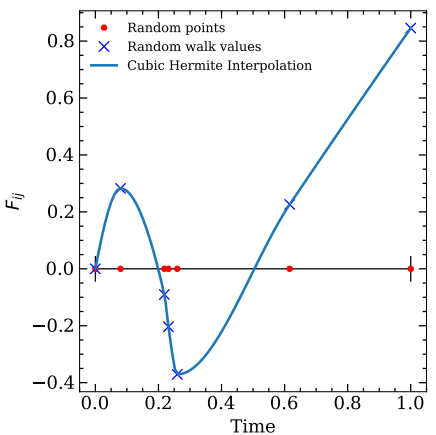
Notational Remark: $F \in \mathbb{R}^6$ as a vector and $\mathbf{F} \in \mathbb{R}^{3\times3}_{sym}$ as a matrix.

UNDERPINNING THEORY

- Conditions for existence of a solution of the macro problem
  - Convexity: $\mathcal{W}(F, \xi) = g(F, \mathrm{adj}\, F, \det F, \xi)$ where $g(\cdot, \cdot, \cdot, \xi)$ is convex.
  - Coercivity: $\mathcal{W}(F, \xi)/|F|^3 \to \infty$ as $|F| \to \infty$.
- Second Law of Thermodynamics
  - $\mathcal{D}$ is convex
  - $\mathcal{D}(\dot{\xi}) \geq c|\dot{\xi}|$
- Material Stability
  - $\mathcal{W}(F, \xi) \to \infty$ as $\det F \to 0^+$
- Frame-indifference
  - $\mathcal{W}(\mathbf{QF}) = \mathcal{W}(\mathbf{F}), \quad \forall\, \mathbf{Q} \in \mathrm{SO}(3)$

Ball, 1977; Ball, 1976; Mielke, 2003; Ortiz and Stainier, 1999

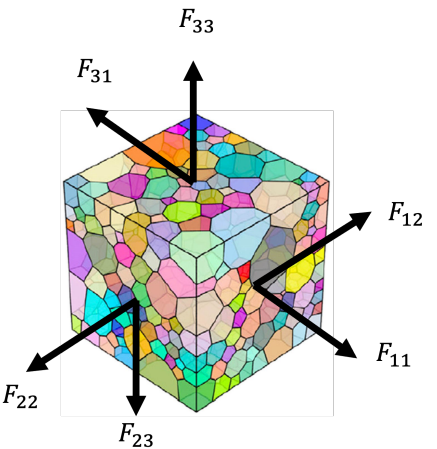
Crystal Plasticity Data



$$(F_{ij})_n = (F_{ij})_{n-1} + \nu F_{max}\sqrt{\Delta t_n}$$

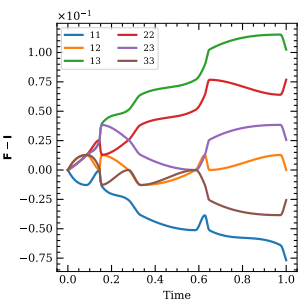
$$\nu \in \{-1, 1\}$$

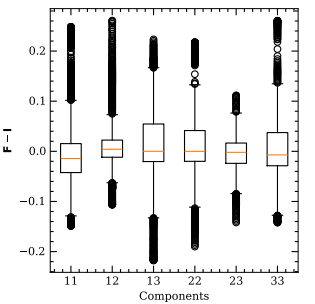
Liu et al., 2023

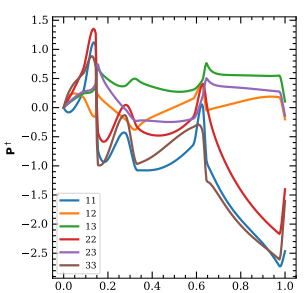


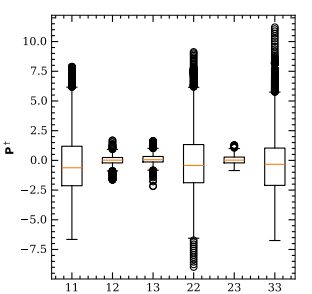
- Crystal plasticity model for each grain.
- Average stress over all the grains in the unit cell (Taylor Averaging).

Data Visualization









Parameterization using Neural Network

Parameterization of $\mathcal{W}$

$$\mathcal{W}(\mathbf{F}, \xi) = \mathcal{N}_H(\det \mathbf{F}) + \mathcal{G}_1(\det \mathbf{F}) + \mathcal{N}_V(\mathbf{F}, \operatorname{adj} \mathbf{F}, \det \mathbf{F}, \xi) + \sum_{ij} \mathcal{G}_2(\mathbf{F}_{ij}) + \mathcal{G}_2(\det \mathbf{F});$$

Parameterization of $\mathcal{D}$

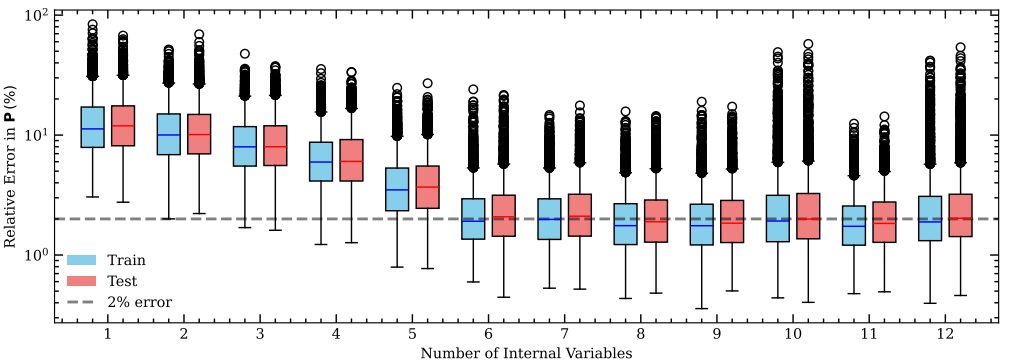
$$\mathcal{D}(\dot{\xi}) = \mathcal{N}_D(\dot{\xi}) + \mathcal{G}_2(\dot{\xi}),$$

Neural Network Architecture

- $\mathcal{N}_H$, $\mathcal{N}_V$ and $\mathcal{N}_D$ are convex by architecture design.
- $\mathcal{G}_1$ and $\mathcal{G}_2$ are used for coercivity of $\mathcal{W}$ and $\mathcal{D}$.

Optimal number of Internal Variables

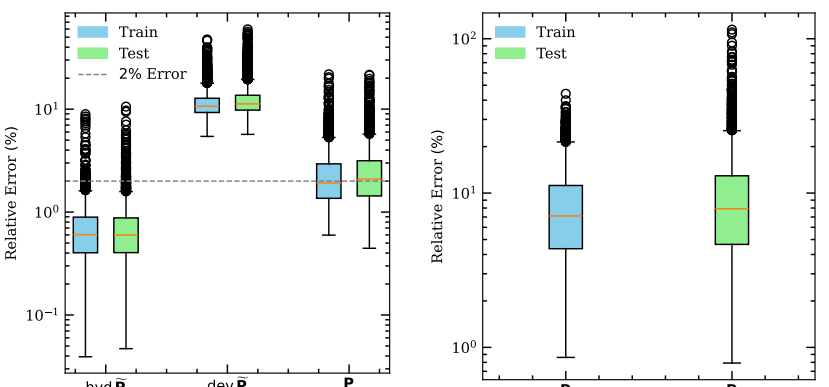
The number of internal variables are unknown apriori!



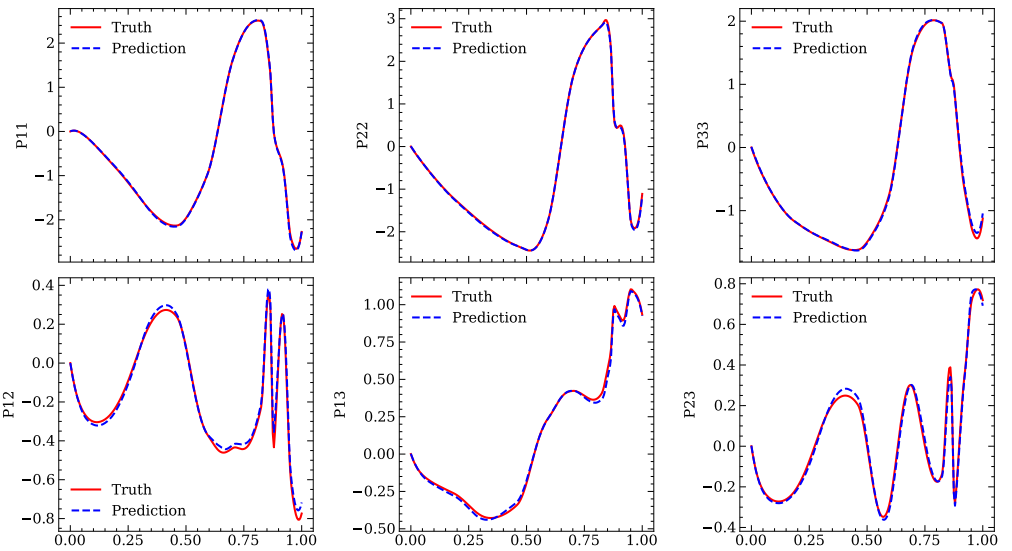
Six is the optimal number of internal variables.

Stress Prediction Accuracy

$$\mathcal{W}(\mathbf{F},\xi) = \mathcal{N}_H(\det \mathbf{F}) + \mathcal{G}_1(\det \mathbf{F}) + \mathcal{N}_V(\mathbf{F}, \operatorname{adj} \mathbf{F}, \det \mathbf{F}, \xi) + \sum_{ij} \mathcal{G}_2(\mathbf{F}_{ij}) + \mathcal{G}_2(\det \mathbf{F});$$

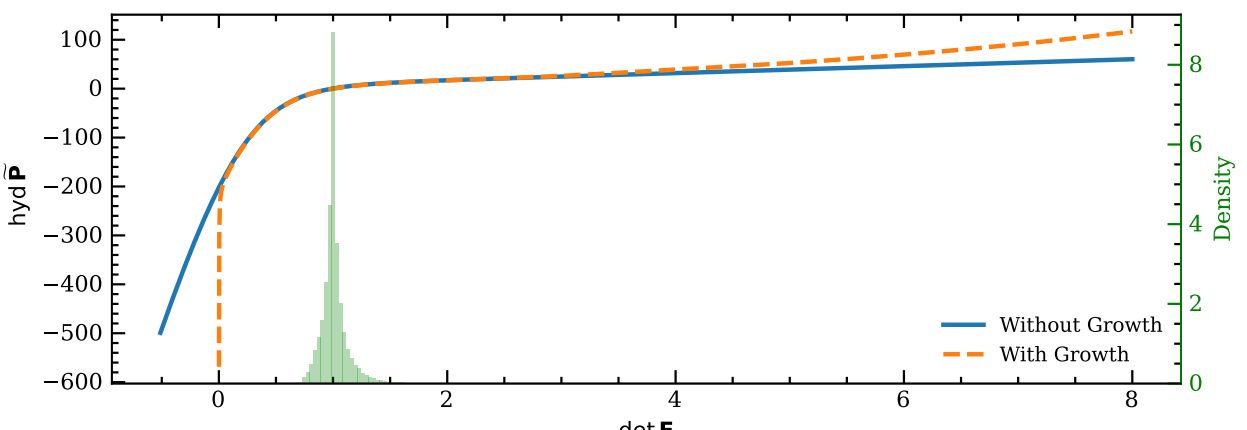


Comparison of True and Predicted Stress



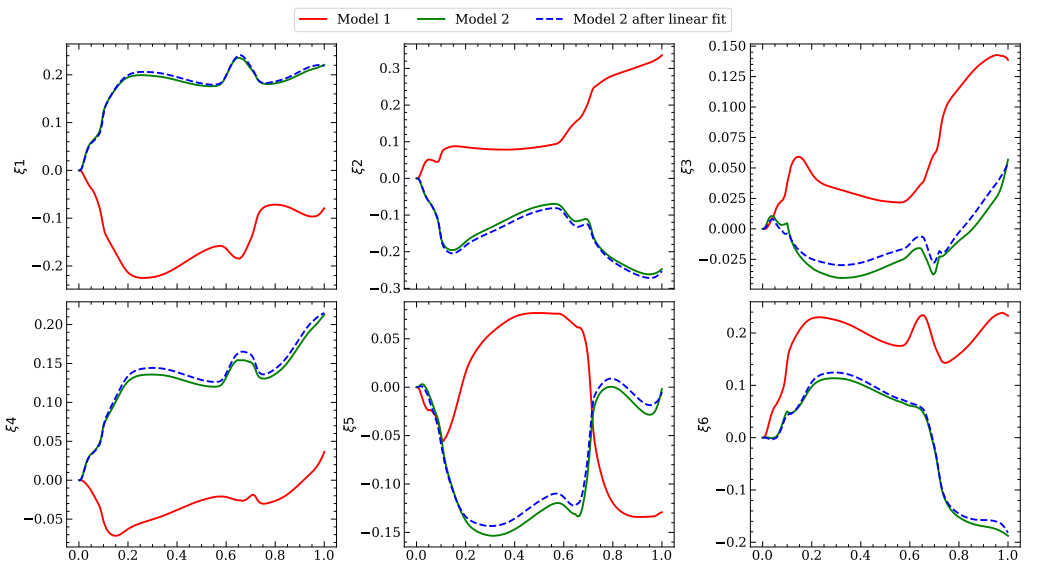
Coercivity

$$\mathcal{W}(\mathbf{F}, \xi) = \mathcal{N}_H(\det \mathbf{F}) + \mathcal{G}_1(\det \mathbf{F}) + \mathcal{N}_V(\mathbf{F}, \operatorname{adj} \mathbf{F}, \det \mathbf{F}, \xi) + \mathcal{G}_2(|\mathbf{F}|) + \mathcal{G}_2(\det \mathbf{F});$$



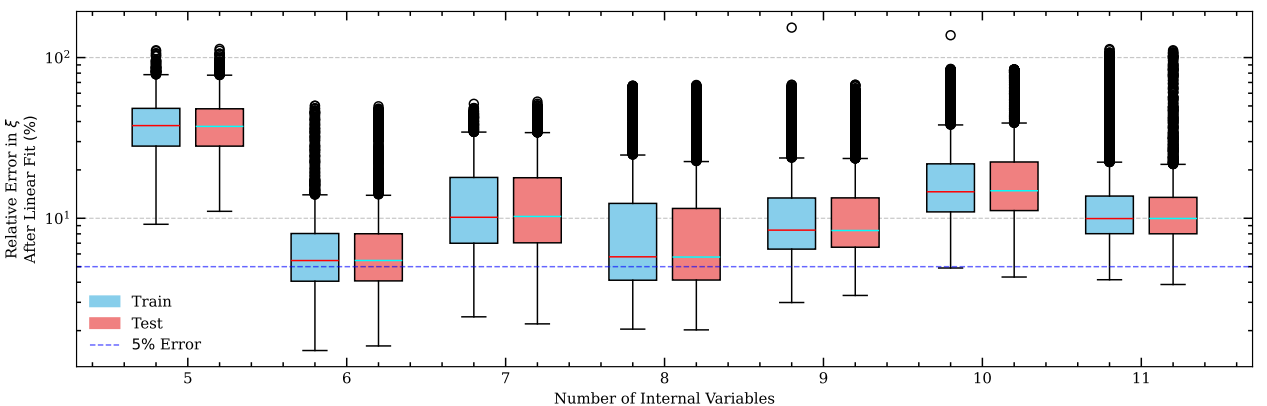
Comparison of Internal Variables

$\xi(t) = A\eta(t)$



Error in the Linear Fit of Internal Variables

$||\xi - A\eta||_{L^2}$



Theory Supporting Identifiability of Internal Variables

Let $F \in \mathbb{R}^m$, $\xi \in \mathbb{R}^n$ and $\eta \in \mathbb{R}^n$, such that $m \geq n$ and consider the two models

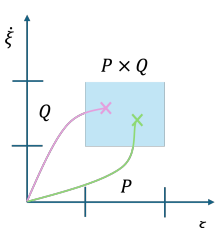
$$\mathcal{W} : (F, \xi) \to \mathcal{W}(F, \xi)$$
$$\mathcal{D} : \dot{\xi} \to \mathcal{D}(\dot{\xi})$$

$$P(t) = \partial_F \mathcal{W}(F(t), \xi(t))$$
$$\partial_{\dot{\xi}} \mathcal{D}(\dot{\xi}) + \partial_\xi \mathcal{W}(F(t), \xi(t)) = 0, \quad \xi(0) = 0.$$

$$\widetilde{\mathcal{W}} : (F, \eta) \to \mathcal{W}(F, \eta)$$
$$\widetilde{\mathcal{D}} : \dot{\eta} \to \mathcal{D}(\dot{\eta})$$

$$P(t) = \partial_F \widetilde{\mathcal{W}}(F(t), \eta(t))$$
$$\partial_{\dot{\eta}} \widetilde{\mathcal{D}}(\dot{\eta}) + \partial_\xi \widetilde{\mathcal{W}}(F(t), \eta(t)) = 0, \quad \xi(0) = 0.$$

Let $\mathcal{T} : \xi \mapsto \eta$ be a diffeomorphism and assume



We can show that $\mathcal{T}|_P$ is linear.

EXAMPLE

Let $m = n$ and define $\mathcal{W}$ and $\widehat{\mathcal{D}}$ as follows

$$\mathcal{W}(F, \xi) = \frac{1}{2} \begin{bmatrix} F \\ \xi \end{bmatrix}^\top \begin{bmatrix} A_1 & A_2 \\ A_2^\top & A_3 \end{bmatrix} \begin{bmatrix} F \\ \xi \end{bmatrix},$$
$$\widehat{\mathcal{D}}(\dot{\xi}) = \frac{1}{2(r+1)} |\dot{\xi}|^{2(r+1)}; \quad \text{where } r > -1/2,$$

where $|\cdot|$ is the 2-norm. Hence, the constitutive law is

$$\left|\dot{\xi}(t)\right|^{2r} \dot{\xi}(t) + A_3 \xi(t) + A_2 F(t) = 0.$$

Hence, for any given t_* and $p, q \in \mathbb{R}^n$, we can construct a polynomial ξ_* such that $\xi_*(t_*) = p$ and $\dot{\xi}(t_*) = q$. We find F_* as

$$F_*(t) = -A_2^{-1} \left(\left|\dot{\xi}_*(t)\right|^{2r} \dot{\xi}_*(t) + A_3 \xi_*(t) \right)$$

Hence, $P = Q = \mathbb{R}^n$.

Frame Indifference

Definition (Requirement)

$$\begin{aligned} \mathcal{W}(\mathbf{QF}) &= \mathcal{W}(\mathbf{F}) \\ \mathbf{P}(\mathbf{QF}) &= \mathbf{QP}(\mathbf{F}), \quad \mathbf{Q} \in \mathrm{SO}(3) \end{aligned}$$

Extension of the Surrogate Model that is Consistent With Frame Indifference

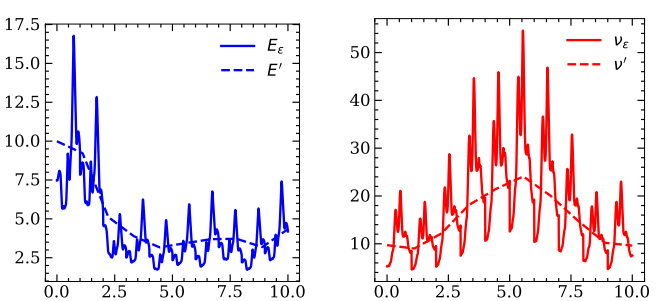
$$\mathbf{P}(\mathbf{F}) := \begin{cases} \mathbf{P}_D(\mathbf{F}), & \text{if } \mathbf{F} \in \mathbb{R}^{3\times 3}_{sym}, \\ \widetilde{\mathbf{Q}}\mathbf{P}_D(\widetilde{\mathbf{F}}), & \text{if } \mathbf{F} \neq \mathbb{R}^{3\times 3}_{sym},\ \mathbf{F} = \widetilde{\mathbf{Q}}\widetilde{\mathbf{F}}, \widetilde{\mathbf{Q}} \in \mathrm{SO}(3),\ \widetilde{\mathbf{F}} \in \mathbb{R}^{3\times 3}_{sym}. \end{cases}$$

Part II

Microstructure-dependent Learning

1D Kelvin-Voigt Viscoelasticity Model (Multiscale Problem)

$$\underbrace{\partial_x \left(E_\varepsilon \overbrace{\partial_x u_\varepsilon}^{F} + \nu_\varepsilon \overbrace{\partial_{tu}^2 u_\varepsilon}^{\dot{F}} \right)}_{P_\varepsilon^\dagger(F,\dot{F})} = 0, \quad (x,t) \in \Omega \times \mathcal{T}, \text{B.C and I.C.}$$



Effective Properties

$$\nu' = \left(\int_{\mathbb{T}} \frac{1}{\nu(y)} \, dy \right)^{-1}$$

$$E' = \frac{\int_{\mathbb{T}} \frac{E(y)}{(\nu(y))^2} \, dy}{\left(\int_{\mathbb{T}} \frac{1}{\nu(y)} dy \right)^2}$$

Bhattacharya et al., 2023; Francfort and Suquet, 1986

Homogenized Kelvin-Voigt Viscoelastic Model (Macro Problem)

$$P^\dagger = E'(x)\underbrace{\partial_x u_0}_{F} + \nu'(x)\underbrace{\partial_{tu}^2 u_0}_{\dot{F}} - \int_0^t \kappa(t-s,x)\underbrace{\partial_x u_0(s)}_{F}\, ds,$$

Markovian Approximation of the Homogenized KV Model

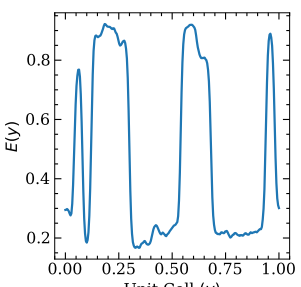
$$P_0^\dagger = E'(x)F + \nu'(x)\dot{F} - \sum_{l=1}^{L} \xi_l,$$
$$\partial_t \xi = \beta_l F - \alpha_l \xi, \quad \xi_l(0) = 0,\ l \in \{1,\dots,L\},$$

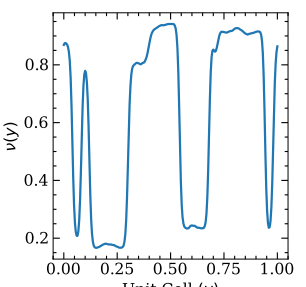
Energy Density and Dissipation

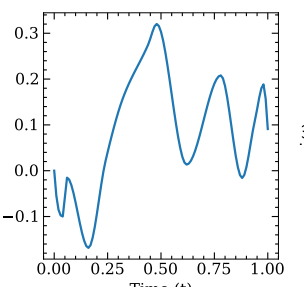
$$\mathcal{W}^\dagger(F,\{\xi_l\}_{l=1}^L,x) = \frac{1}{2}E'(x)F^2 - F\sum_{l=1}^{L}\xi_l + \sum_{l=1}^{L}\frac{\alpha_l(x)}{2\beta_l(x)}\xi_l^2$$
$$\mathcal{D}^\dagger(\dot{F},\{\dot{\xi}_l\}_{l=1}^L,x) = \frac{1}{2}\nu'(x)\dot{F}^2 + \sum_{l=1}^{L}\frac{1}{2}\frac{\dot{\xi}_l^2}{\beta_l(x)};$$

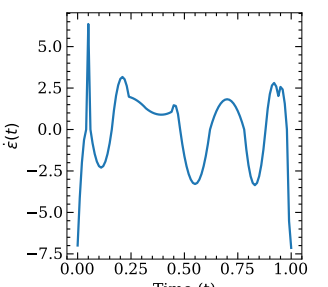
A Sample from the Dataset

$\{E, \nu, F, \dot{F}\}$









Neural Network Architecture

Energy Function Architecture

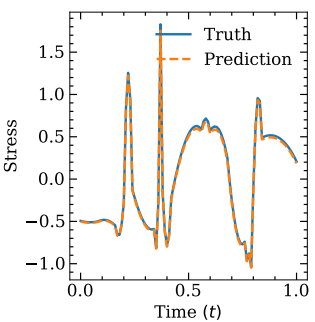
$$\overbrace{\phi_1}^{\textit{vector}} = \mathcal{N}_1(\overbrace{E, \nu}^{\text{Functions}}),$$
$$\mathcal{W} = \mathcal{N}_2(\underbrace{F, \{\xi_l\}_{l=1}^{L}, \phi_1}_{\textit{vectors}}),$$

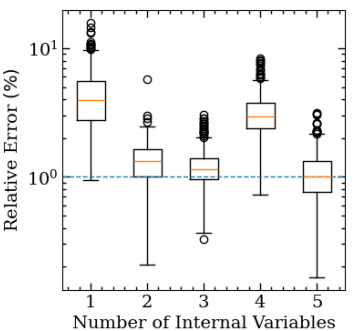
Dissipation Function Architecture

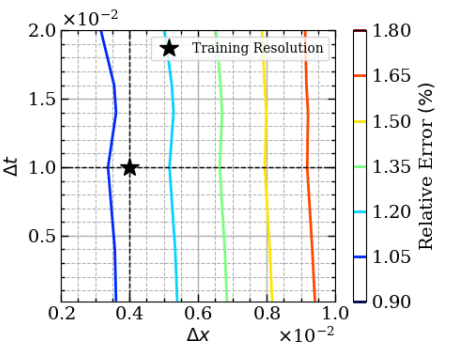
$$\phi_2 = \mathcal{N}_3(E, \nu);$$
$$\phi_3 = \mathcal{N}_4(E, \nu);$$
$$\mathcal{D} = -\mathcal{N}_5(\dot{F}, \phi_2) + \mathcal{N}_6(\{\dot{\xi}_l\}_{l=1}^{L}, \phi_3),$$

$\mathcal{N}_{1,3,4}$: Fourier Neural Mappings. These networks act like encoder that map microstructure to finite-dimensional representation!
$\mathcal{N}_{2,5,6}$: Convex Architecture

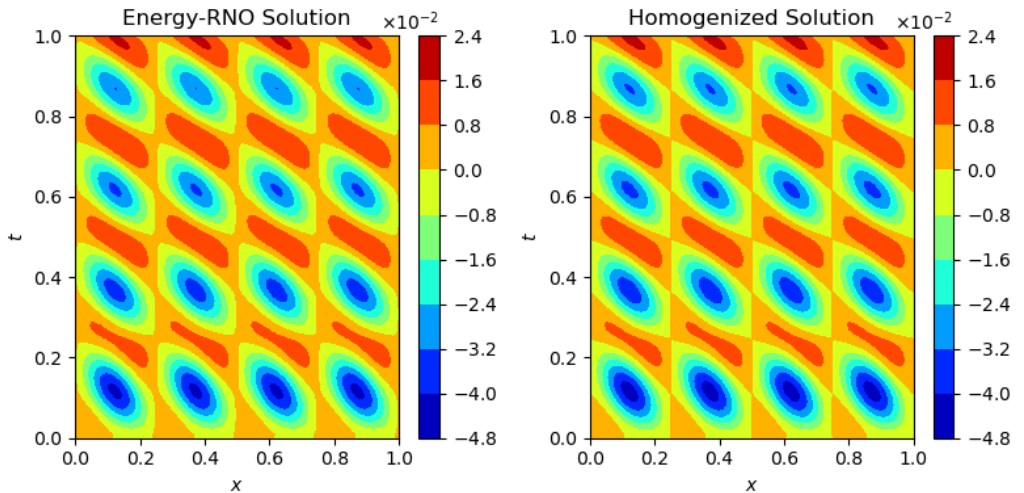
Stress Prediction Accuracy





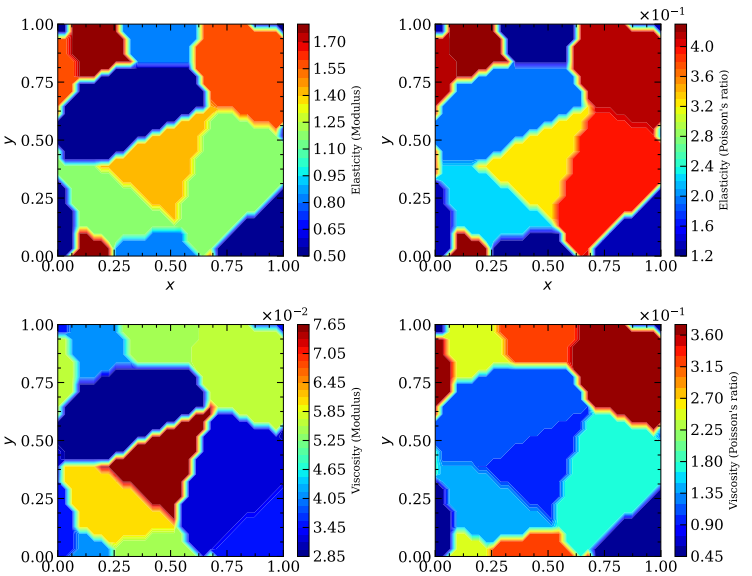


Comparison of Surrogate Model with Multiscale Solution

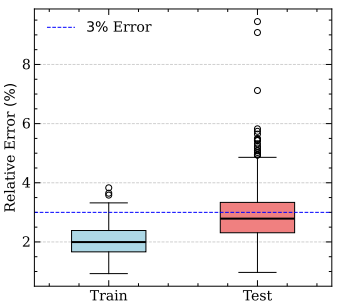


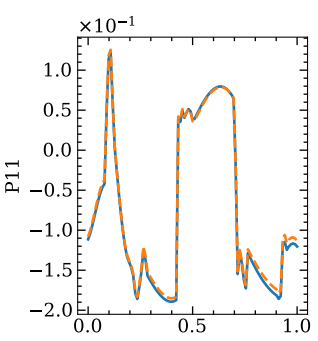
Two-Dimensional Kelvin-Voigt Multiscale Model

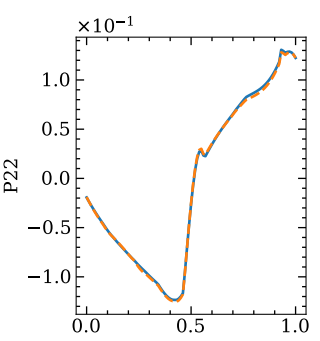
$$\nabla \cdot (E_\varepsilon \nabla u_\varepsilon + \nu_\varepsilon \nabla \partial_t u_\varepsilon) = 0, \quad (x, t) \in \Omega \times \mathcal{T}, \text{B.C and I.C.}$$

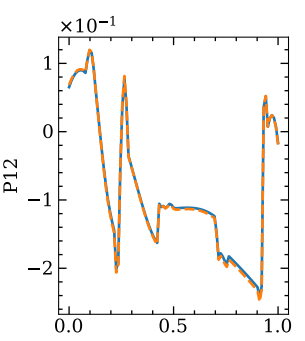


Stress Prediction Accuracy

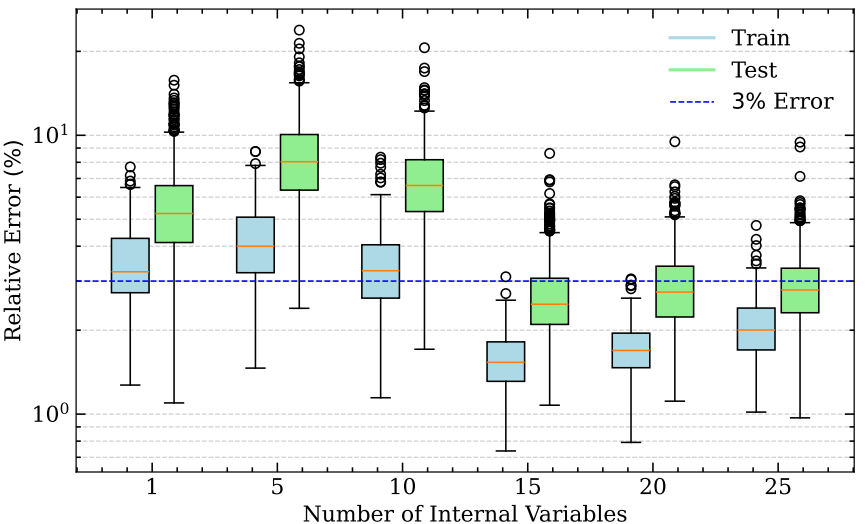






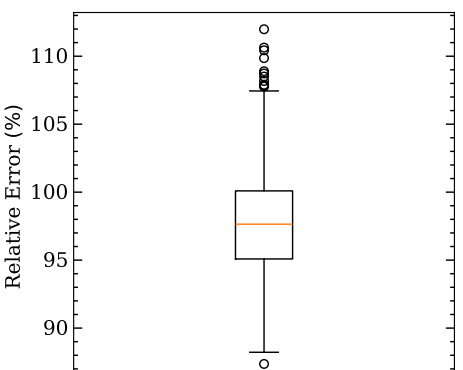


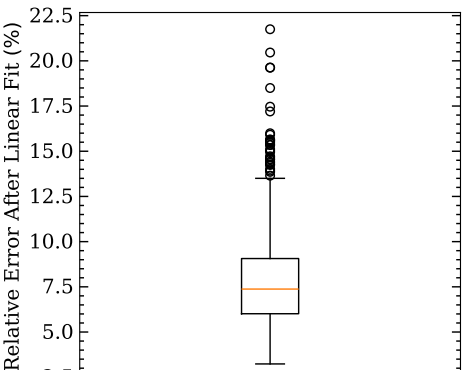
Number of Internal Variables



Identifiability of Internal Variables up-to linear transform

$\xi, \eta \in \mathbb{R}^{25}, \quad \xi(t) = A\eta(t)$





TEMPORALLY NON-LINEAR MODEL- VISCOPLASTICITY

Multiscale Model

$$P_\varepsilon(x,t) = E_\varepsilon(x)(F_\varepsilon(x) - \xi_\varepsilon(x)),$$
$$\dot{\xi}_\varepsilon(x,t) = c_{0\varepsilon}(x)\mathrm{sign}(P_\varepsilon(x,t)) \left(\frac{|P_\varepsilon(x,t)|}{P_{0\varepsilon}(x)}\right)^{n_\varepsilon(x)}$$

Homogenized Model

$$\mathcal{W}^\dagger(F,\xi,x) = \frac{1}{2}E'(x)(F-\xi)^2,$$
$$\mathcal{D}^{\dagger *}(q,x) = \int_{\mathbb{T}} c_0(x,y)P_0(y)^{-n(x,y)} \left(\frac{1}{n(x,y)+1}|q|^{n(x,y)+1}\right) dy.$$

Neural Network Architecture

Energy Function Architecture

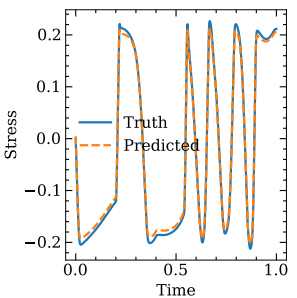
$$\phi_1 = \mathcal{N}_1(E),$$
$$\mathcal{W} = \mathcal{N}_2(F, \xi, \phi_1),$$

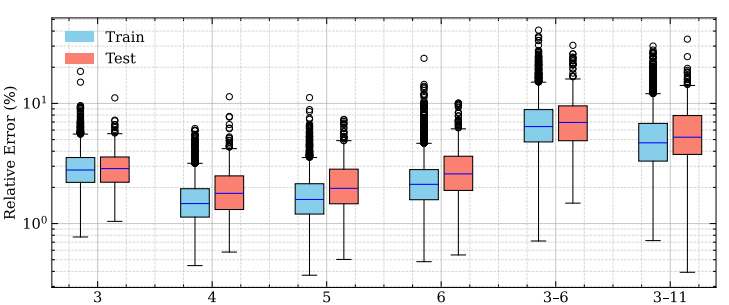
Dissipation Function Architecture

$$\phi_2 = \mathcal{N}_3(c_0, P_0, n);$$
$$\mathcal{D} = \mathcal{N}_4(\{\dot{\xi}_l\}_{l=1}^{L}, \phi_2),$$

$\mathcal{N}_{1,6}$: Fourier Neural Mappings. These networks act like encoder that map microstructure to finite-dimensional representation!
$\mathcal{N}_{2,4}$: Convex Architecture

Predictive Accuracy





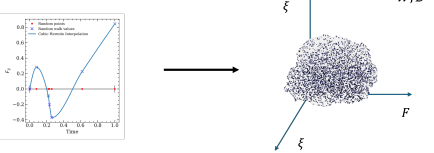
CONTRIBUTIONS

- A framework for learning material models that is consistent with the second law of thermodynamics and the theory underpinning the well-posedness and stability of the conservation equation.
- Identifiability of internal variables up to linear transform and supporting theory.
- The framework is deployed and demonstrated to work in various settings: crystal plasticity, Kelvin-Voigt viscoelasticity.
- The framework is extended to learn microstructure-dependent constitutive laws.

Raj, M., Cao, L., Stuart, A., & Bhattacharya, K. (2026). **A neural-network framework to learn history-dependent constitutive laws and identifiability of internal variables.**

Future Work

- Use the machine-learning closure model in a conservation law and do a computational cost comparison.
- Optimal method for sampling data.



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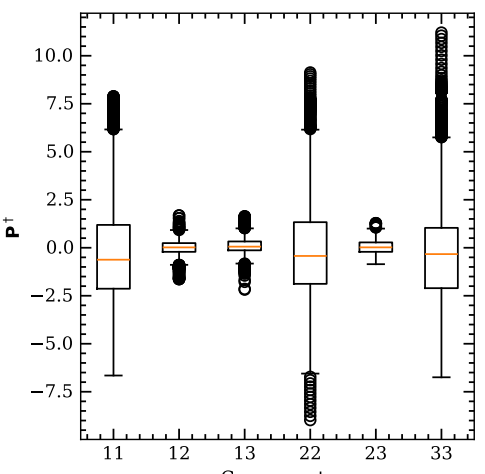
Part III

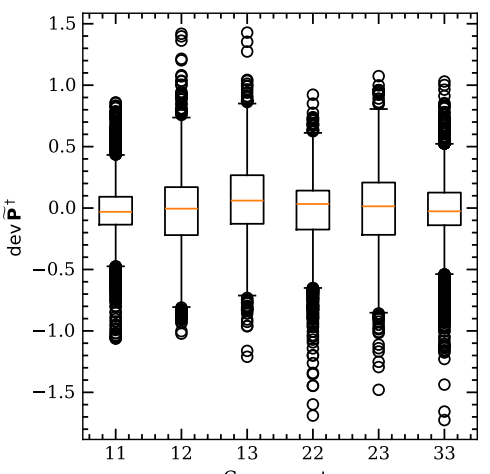
APPENDIX

Hydrostatic Deviatoric Decomposition

$$\mathcal{W}(\mathbf{F}, \xi) = \mathcal{W}_H(I_F) + \mathcal{W}_V(\mathbf{K}, \xi) \tag{2}$$

where $I_F = \det \mathbf{F}$ and $\mathbf{K} := \mathbf{F}/I_F^{1/3}$; $\mathcal{W}_H : \mathbb{R} \to \mathbb{R}$, $\mathcal{W}_V : \mathbb{R}^{d\times d} \times \mathbb{R}^n \to \mathbb{R}$.





Convex Architecture

$$h_1 = \mathrm{ReLUS}\left(W_1 h_0 + b_1\right);$$
$$h_2 = W_2 h_1 + b_2.$$

where

$$\mathrm{ReLUS}(x) = \begin{cases} 0 & \text{if } x \leq 0, \\ x^2 & \text{if } x > 0. \end{cases}$$

and $W_2 = \widetilde{W_2} \odot \widetilde{W_2}$.

Amos et al., 2017

Functions to Impose Growth

